

Pavan B N

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LinkedIn | GitHub | LeetCode

WORK EXPERIENCE

IBM

Application Developer

Bengaluru, India

Jun 2026 – Present

- Develop **2 Spring Boot microservices** and their **REST APIs** within an application modernization and migration program, delivering full-stack features across **Java, Spring Boot, and React**.
- Reduced API response time from **200ms to 20ms** by introducing **Redis** caching for frequently accessed data, and integrated **Apache Kafka** across **5 topics** to decouple services through asynchronous, event-driven processing.
- Own end-to-end deployment for both services by authoring **Dockerfiles** and **Kubernetes** manifests, enabling containerized, repeatable delivery to the cluster.

Microsoft

Software Engineer Apprentice

Bengaluru, India

Nov 2025 – Jun 2026

- Redesigned iteration-run retrieval for Microsoft Fabric FMLV **REST APIs**, replacing repeated File System initialization with batch initialization and map-based lookups, reducing lookup complexity from **O(n)** to **O(1)** and improving scalability as iteration counts grew.
- Remediated **70+ security vulnerabilities (CVEs)** in Apache Ambari by tracing transitive-dependency chains, upgrading affected packages, and validating compatibility across a legacy codebase with **zero functional regressions**.

Blink Health

Automation Engineer Intern

Bengaluru, India

Jul 2024 – Jun 2025

- Replaced browser-driven E2E validation with a reusable **HTTP client automation framework** across **100+ API endpoints**, raising automation reliability from **70% to 100%** and cutting execution time by **40%**.

PROJECTS

Readora (Cloud-Native, AI-Powered Bookstore Platform)

[GitHub](#)

- Architected an event-driven platform of **10 Spring Boot microservices** behind **Spring Cloud Gateway** with JWT authentication and Redis-backed rate limiting; coordinated **10 Kafka topics** using the transactional **outbox pattern** with idempotent consumers, circuit breakers, Redis caching, and **Flyway**-managed migrations.
- Built an **MCP server** exposing catalog, cart, and order data as tool-callable endpoints for a **RAG** reading assistant (**Azure OpenAI** + pgvector semantic search); shipped **3 React + TypeScript** applications using **Redux Toolkit** and **STOMP/SockJS** for real-time order and notification updates.
- Delivered a **GitOps** pipeline of **13 GitHub Actions** workflows that enforce a **90% JaCoCo coverage gate** (JUnit 5, **Testcontainers**), security-scan images, and push to **Azure Container Registry**, while **ArgoCD** auto-syncs an **AKS** cluster with **OpenTelemetry** distributed tracing to New Relic.

TECHNICAL SKILLS

Languages: Java, JavaScript, TypeScript, SQL

Backend: Spring Boot, REST APIs, Microservices, Apache Kafka, Redis

Frontend: React.js, Redux, HTML5, CSS3

Databases: PostgreSQL, MySQL

Cloud & DevOps: Azure, Docker, Kubernetes, Git, GitHub Actions, CI/CD

AI: LLMs, RAG, Agentic AI, Model Context Protocol (MCP), Azure OpenAI

Core CS: Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks, Distributed Systems, System Design

EDUCATION

Reva University

B.Tech in Computer Science and Engineering (GPA: 9.16 / 10)

Bengaluru, India

Oct 2021 – May 2025

ACHIEVEMENTS & CERTIFICATIONS

- Solved **500+ DSA problems** across **LeetCode**, **Codeforces**, **GeeksforGeeks**, and **Take U Forward (TUF)**, covering Graphs, Dynamic Programming, Trees, Greedy Algorithms, and Binary Search.
- Built and showcased solutions in the **Microsoft Agent League Hackathon**, **Smart India Hackathon (SIH)**, **CodeFury**, and **Reva Hack**, across Agentic AI, Multi-Agent Systems, and Cloud-Native Applications.
- Certified in **Microsoft Azure Fundamentals (AZ-900)** and **Microsoft Azure AI Fundamentals (AI-900)**.